

Geometry-MMALS G1

Status, Evidence, Limitations, and Research Perspective

After the v1.0.9 Five-Seed Stationary-Geometry Pilot

Reviewer-oriented arXiv-style technical report, version 1.2

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Evidence snapshot: Geometry-MMALS G1 v1.0.9, five model seeds, matched-compute development
protocol
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Epistemic status. This report documents the first replicated candidate result of the Geometry-MMALS G1 program. Across five matched-compute model seeds, the complete four-dimensional context-alignment objective improves trained-angle context distance-order correlation and reduces context stress relative to a no-geometry control. It also improves factor decoding on held-out source identities. These effects do not extend reliably to route geometry, synthesis geometry, predictive accuracy, forgetting, or causal specificity. The stationary route target is theoretically cleaner and numerically stable, but it does not outperform the legacy route target in this pilot. The report therefore supports a narrow candidate C1 context-geometry claim, not a completed claim of functional geometry, adaptive-route superiority, host specialization, continual memory transport, or operational utility. No claim of quantum computation or quantum advantage is made.

Abstract

Geometry-MMALS G1 asks whether a continual-learning system composed of a frozen sensory representation, inferred context, adaptive routing, functional hosts, and memory can acquire an internal geometry that is observable, predictive, causally controllable, and operationally useful. The program distinguishes internal functional geometry from the external geometry of audit variables and uses a staged evidence ladder from pipeline integrity to descriptive order, interpolation, specialization, causal control, continual transport, and operational utility.

This report updates the reviewer-facing synthesis through the completed v1.0.9 five-seed pilot. The principal result is replicated but narrow. Relative to a matched no-geometry control, the four-dimensional globally aligned objective increases trained-angle context distance-order correlation by 0.107 with a 95% seed interval of [0.029, 0.185], while context stress decreases by 0.042 with interval [−0.084, −0.001]. All five seeds show a positive context-order effect. On held-out source identities, context-factor decoding improves by 0.132 in R^2 and angular error decreases by approximately 5° . In contrast, route-order effects are not replicated, synthesis geometry remains unchanged, held-out and final accuracy are statistically neutral, forgetting is unchanged, and causal specificity remains below the preregistered threshold. The stationary route target does not outperform the legacy target and slightly worsens synthesis geometry.

The evidence therefore supports a reviewer-safe statement: Geometry-MMALS can be trained to learn a reproducibly more ordered and more decodable latent context representation on a controlled rotation benchmark, but the current architecture does not yet convert that representation into functional control. The next research priority is no longer to increase context correlation in isolation, but to establish a causal and operational bridge from context geometry to routes, host transformations, synthesis, and continual retention.

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1 Purpose and scope

This document is not a replacement for the original Geometry-MMALS G1 specification. It is a status and perspective report for reviewers, collaborators, and future contributors. It has five purposes:

1. state the strongest claim currently supported by the evidence;
2. separate replicated positive results from single-seed observations and failed gates;
3. explain the mathematical and experimental revisions from v1.0.0 through v1.0.9;
4. expose the main reviewer risks before an empirical submission;
5. define the shortest defensible path from representational geometry to functional geometry.

One-sentence status. Across five matched-compute RotatedMNIST runs, a four-dimensional globally aligned objective reproducibly improves the geometry and held-out decodability of the inferred context, but it does not improve route geometry, synthesis geometry, prediction, forgetting, or causal specificity.

2 Research question and conceptual contribution

2.1 From audit geometry to functional geometry

A vector of measured quantities such as accuracy, cost, forgetting, drift, and route entropy can be embedded and analyzed geometrically. That is useful for audit and control, but it does not demonstrate that the learning system itself acquired a meaningful internal geometry. Geometry-MMALS G1 therefore studies the internal chain

$$x \longrightarrow z_0 \longrightarrow c \longrightarrow r \longrightarrow \{g_h(z_0)\}_{h=1}^H \longrightarrow z_M \longrightarrow \hat{y}, \quad (1)$$

where z_0 is the frozen sensory representation, c is inferred context, r is a route on the probability simplex, g_h is host h , z_M is the routed synthesis, and $\hat{y}$ is the prediction.

The scientific question is not whether these objects can be plotted. It is whether their distances, neighborhoods, trajectories, interventions, and temporal transport correspond to a known physical factor in ways that survive controls, held-out source identities, model seeds, and operational tests.

2.2 Why the problem is geometric rather than metaphorical

A geometric claim becomes scientifically meaningful when it predicts or constrains relations. In the controlled rotation setting, neighboring angles should induce neighboring internal states; unseen intermediate angles should occupy ordered positions; a latent direction associated with increasing rotation should produce a consistent downstream change; and continual updates should preserve or controllably deform previously learned relations. This framing connects manifold learning [1, 2], information geometry [3], geometric deep learning [4], and representation comparison [5, 6] to a modular continual-learning system.

2.3 Philosophical status

The report adopts a constrained structural realism: internal coordinates are not assumed to reveal an intrinsic ontology, but stable relations may still be real enough to support prediction, intervention, and engineering. A latent geometry is therefore accepted only to the degree that it survives a hierarchy of tests. This stance is compatible with structural realism [7] and interventionist accounts of causal explanation [8].

2.4 Potential novelty

The potential contribution is not merely another regularized latent space. The intended novelty is the joint qualification of:

- a context geometry grounded in a controlled factor;
- a route geometry on the simplex;
- host functions whose roles are spatially and causally localized;
- memory as preservation or controlled transport of functional relations;
- a claim manifest linking every statement to auditable evidence.

The v1.0.9 result establishes only the first item at candidate level.

3 Mathematical model

3.1 Sensory representation, context, route, and synthesis

The frozen encoder produces

$$z_0 = P(x) \in \mathbb{R}^{d_z}. \quad (2)$$

The context encoder produces a raw context $c_{\text{raw}} = C(z_0)$, which is normalized before use by the context-bottleneck router:

$$c = \frac{c_{\text{raw}}}{\max(\|c_{\text{raw}}\|_2, \varepsilon)}. \quad (3)$$

The route is

$$r = \text{softmax } R_c(c), \quad r \in \Delta^{H-1}. \quad (4)$$

The host synthesis is

$$z_M = \sum_{h=1}^H r_h g_h(z_0), \quad \hat{y} = K(z_M). \quad (5)$$

The current protocol is G1-A: the known rotation angle is used to shape geometry during training, but it is not supplied to the deployed router. This is a controlled grounding study, not unsupervised manifold discovery.

3.2 Direct context geometry

For the same source identity s observed at factors u_a and u_b , the half-chord context distance is

$$d_C(c_{s,a}, c_{s,b}) = \frac{1}{2} \|c_{s,a} - c_{s,b}\|_2. \quad (6)$$

The fixed chord-compatible target over the 120° training span is

$$\tau(u_a, u_b) = \sin \left[\frac{\pi}{2} \min \left(\frac{|u_a - u_b|}{120^\circ}, 1 \right) \right]. \quad (7)$$

The local context objective is

$$\mathcal{L}_{\text{context}} = \mathbb{E}_s \sum_{a < b} [d_C(c_{s,a}, c_{s,b}) - \tau(u_a, u_b)]^2 + \lambda_{\text{far}} \mathcal{L}_{\text{far}} + \lambda_{\text{spread}} \mathcal{L}_{\text{spread}}. \quad (8)$$

The far-pair term prevents distant angles from remaining too close. The path-spread term prevents each source trajectory from becoming nearly constant. v1.0.8 showed that these local terms alone were insufficient; local trajectories could move without sharing a common orientation.

3.3 Cross-source fiber alignment

For adjacent ordered angles, define the source-specific transition

$$\Delta c_{s,a} = c_{s,a+1} - c_{s,a}. \quad (9)$$

Fiber alignment encourages transitions over the same factor interval to share a direction across source identities:

$$\mathcal{L}_{\text{fiber}} = \frac{1}{A-1} \sum_a \mathbb{E}_s \left[1 - \cos(\Delta c_{s,a}, \bar{\Delta c}_a) \right], \quad (10)$$

where $\bar{\Delta c}_a$ is the mean transition for interval a . A centroid grounding term additionally constrains factor-level centroids. This identifies global structure only up to an isometry; signed factor decoding and causal interventions test functional orientation.

3.4 Route geometry: legacy and stationary targets

Routes are compared in square-root-simplex coordinates. The numerically stable stationary distance is

$$d_R(p, q) = \frac{\|\sqrt{p} - \sqrt{q}\|_2}{\sqrt{2}}. \quad (11)$$

The legacy objective normalized factor distances by the maximum gap visible at the current continual stage, so the target for a fixed pair changed as the curriculum expanded. v1.0.9 replaced it with the same fixed target $\tau(u_a, u_b)$ used by context geometry. This is theoretically preferable because the target is independent of stage and curriculum span.

Two execution-stability corrections were required before the pilot completed: deterministic bootstrap seeds were mapped into the non-negative uint32 domain, and $\sqrt{1 - \langle \sqrt{p}, \sqrt{q} \rangle}$ was replaced by the algebraically equivalent root-space norm above to avoid a singular gradient for coincident routes.

3.5 Complete training objective

The aligned model optimizes

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_{\text{anchor}} \mathcal{L}_{\text{anchor}} + \lambda_r \mathcal{L}_{\text{route}} + \lambda_c \mathcal{L}_{\text{context}} + \lambda_f \mathcal{L}_{\text{fiber}} \quad (12)$$

$$+ \lambda_g \mathcal{L}_{\text{centroid}} + \lambda_h \mathcal{L}_{\text{host-div}}. \quad (13)$$

The retention anchor is controlled rehearsal over previously seen transformed views. It is not yet the reconstructive or synthetic dual memory envisioned by MMALS.

3.6 Causal probe

A factor direction is fitted on one subset of source identities and evaluated on a disjoint subset. For an intervention scale α ,

$$c' = \frac{c + \alpha v_u}{\|c + \alpha v_u\|_2}, \quad (14)$$

and the downstream route effect is compared with matched orthogonal directions. The causal-specificity ratio is

$$\text{CSR} = \frac{|\Delta r_{v_u}|}{\mathbb{E}_j |\Delta r_{v_j^\perp}| + \varepsilon}. \quad (15)$$

A candidate C4 result requires orientation, monotonicity, class-evidence preservation, and a CSR lower confidence bound above one; the pilot threshold for mean specificity is 1.5.

4 Evidence ladder and claim discipline

Level	Claim family	Required evidence
C0	Pipeline integrity	Executable package, finite losses and gradients, deterministic manifests, equal-compute controls, version and source hashes.
C1	Descriptive geometry	Factor-distance order and stress, held-out source decoding, anti-collapse diagnostics, cross-seed effects.
C2	Predictive geometry	Interpolation and extrapolation benefit beyond matched controls, calibration, paired uncertainty.
C3	Host specialization	Local functional ablation, route-function consistency, cross-seed role matching, stable ecology.
C4	Causal geometry	Source-disjoint interventions, orientation, monotonicity, specificity, identity preservation.
C5	Continual transport	Stage-by-stage preservation or controlled deformation, reconstructive and synthetic memory tests.
C6	Operational utility	Accuracy, retention, cost, stability, or audit benefit attributable to geometry.

A successful C1 result does not imply C2-C6. This separation is the central claim-discipline rule of the program.

5 Research history: what each revision corrected

Revision	Main scientific correction
v1.0.0–1.0.2	Distinguished audit geometry from internal functional geometry; repaired package and numerical route-loss failures.
v1.0.3	Introduced same-source cross-angle batches and source-block metrics, making geometric supervision meaningful.
v1.0.4	Matched initialization and compute; showed that geometry changed route organization but retention anchoring drove practical gains.
v1.0.5	Tested context mediation and curriculum order; exposed direct sensory shortcut and strong order sensitivity.
v1.0.6	Added static route controls and same-final-task counterbalanced curricula; adaptive routing remained only marginally useful.
v1.0.7	Forced context-only routing; established that inferred context was sufficient when the sensory shortcut was removed, but route geometry did not organize context coordinates.
v1.0.8	Added direct context geometry, path spread, fiber alignment, centroid grounding, and dimension-4 capacity; single-seed context geometry became strong but functional gains remained absent.
v1.0.9	Fixed route-target nonstationarity, added dense evaluation and five model seeds, and produced the first replicated candidate context-geometry result.

6 Current experiment: v1.0.9 five-seed pilot

6.1 Controlled benchmark and fixed elements

The pilot uses RotatedMNIST with trained factors $-60, -30, 0, 60, 30$ degrees under a fixed continual curriculum. A frozen sensory grove is trained once and passes a 0° acceptance gate with approximately 94.2% accuracy. Data splits, tangent-fit identities, tangent-evaluation identities, sensory weights, architecture, and compute budgets are fixed across methods. Model initialization varies across seeds 0–4.

Every method uses 15,360 image forwards and 80 optimizer steps per seed. Source-block bootstrap quantifies within-seed uncertainty; Student- t intervals across the five model seeds quantify replication uncertainty.

6.2 Three matched methods

Table 3: v1.0.9 method set.

Method	Context/fiber alignment	Route target	Purpose
d4_no_geo	No	None	Matched no-geometry control
d4_full_legacy_route	Yes	Stage-dependent legacy	Full-alignment reference
d4_full_stationary_route	Yes	Fixed chord-compatible	Primary stationary method

7 Replicated empirical status

7.1 Context geometry replicates

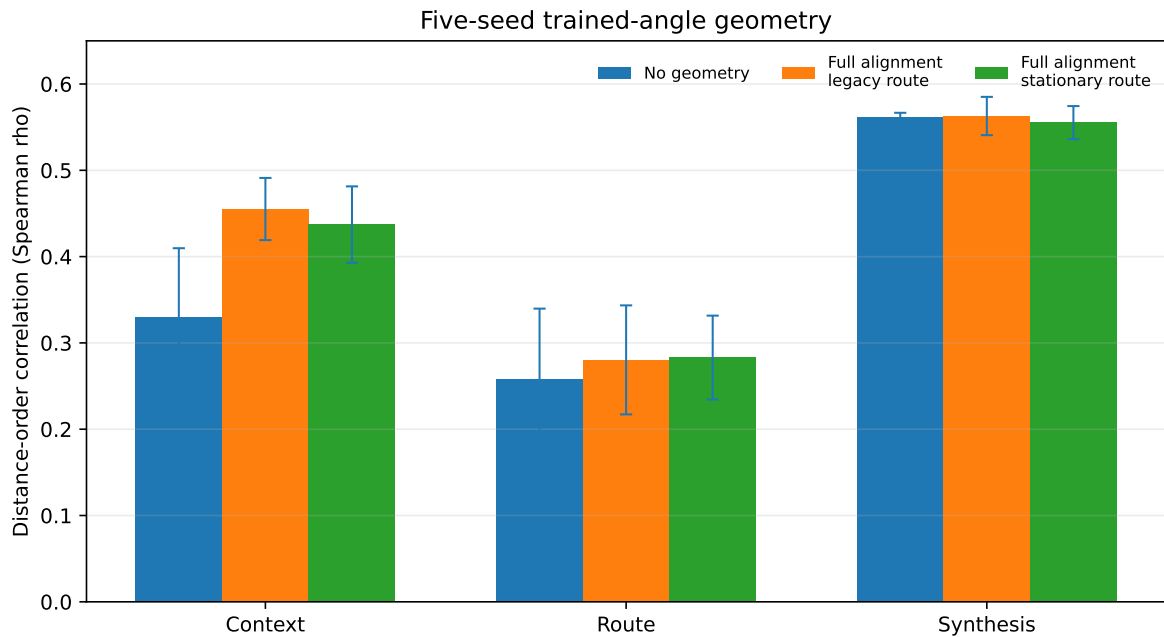


Figure 1: Five-seed trained-angle geometry. Error bars are 95% intervals across model seeds. Context geometry improves under both full-alignment methods; route and synthesis effects are much weaker.

The stationary full-alignment model reaches mean context correlation 0.437 versus 0.330 for the no-geometry control. The paired seed effect is

$$\Delta\rho_{\text{context}} = +0.107, \quad 95\% \text{ CI} = [+0.029, +0.185]. \quad (16)$$

Context stress decreases by 0.042 with interval $[-0.084, -0.001]$. All five seeds have a positive context-order effect. This is the strongest result presently available.

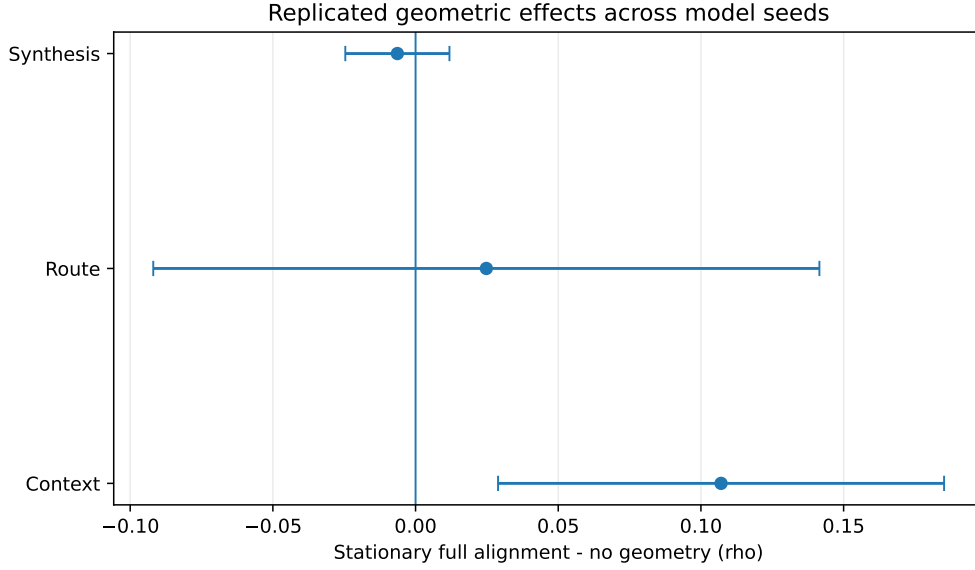


Figure 2: Stationary full alignment minus no geometry. Only the context-order interval excludes zero.

Reviewer-safe positive claim. Across five matched-compute model seeds, the d4 aligned objective reproducibly increases trained-angle context distance-order correlation and reduces context stress on the controlled RotatedMNIST protocol.

7.2 Held-out source factor decoding improves

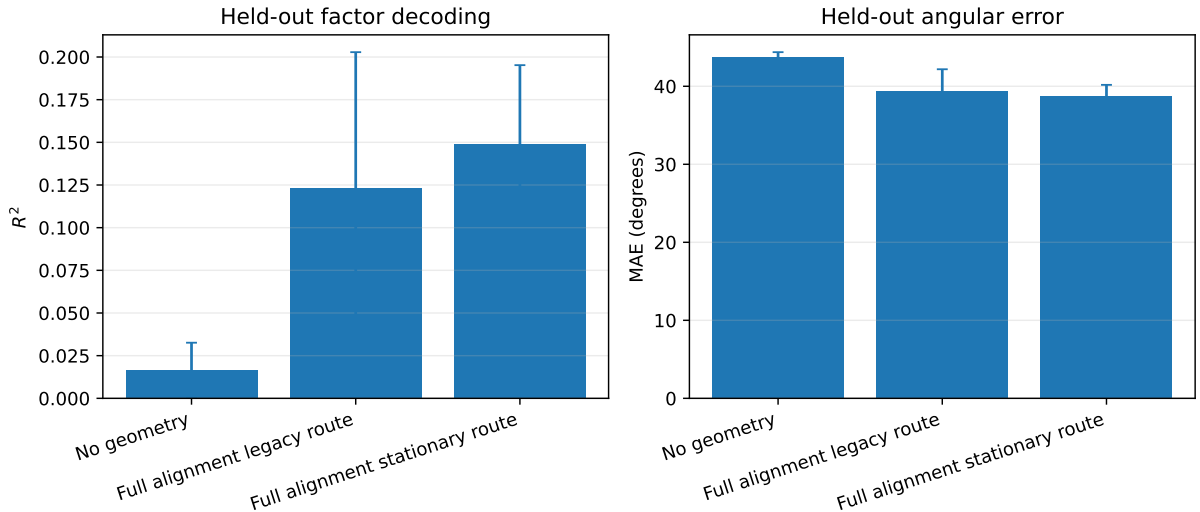


Figure 3: Factor decoding on source identities excluded from tangent fitting. Full alignment improves context R^2 and reduces angular error, while the stationary and legacy route targets are statistically similar.

The stationary model improves held-out context R^2 from 0.016 to 0.149. The paired effect is +0.132 with seed interval $[+0.093, +0.172]$. Mean angular error falls from 43.67° to 38.68° , a reduction of approximately 4.99° with interval $[-6.35^\circ, -3.63^\circ]$.

This result is important because evaluation uses unseen source identities. It argues against an interpretation in which the context merely memorizes source-specific trajectories.

7.3 The stationary route target does not outperform the legacy target

The direct stationary-minus-legacy effects are statistically neutral for context and route geometry:

Metric	Mean effect	95% lower	95% upper
Context ρ	−0.018	−0.045	+0.009
Route ρ	+0.003	−0.034	+0.039
Held-out context R^2	+0.026	−0.037	+0.088
Held-out accuracy	−0.0005	−0.0077	+0.0067
Final trained accuracy	−0.0006	−0.0042	+0.0030
Forgetting	+0.0006	−0.0028	+0.0040

The stationary target is still preferable as a specification because it is curriculum independent and numerically stable. However, empirical superiority is not supported. The replicated context result must be attributed to the broader d4 alignment package, not to the target replacement alone.

7.4 Route geometry does not replicate

The stationary route-order effect versus no geometry is +0.025 with interval $[-0.092, +0.142]$. Only two of five seeds are positive. On the dense held-out angle grid, the effect is −0.006 with interval $[-0.104, +0.093]$.

The five-seed evidence changes the interpretation of v1.0.8. Context geometry replicates; route geometry does not. The program currently has evidence for an ordered latent context representation, not a stable route manifold.

7.5 Synthesis geometry remains unchanged

Stationary full alignment changes synthesis ρ by −0.006 relative to no geometry, with interval $[-0.025, +0.012]$. Relative to the legacy route target, stationary routing reduces synthesis correlation by 0.0076 with a fully negative interval and slightly worsens synthesis stress in every seed.

The desired chain

$$\text{context geometry} \rightarrow \text{route geometry} \rightarrow \text{host synthesis geometry} \quad (17)$$

is therefore not established. The current result remains representational rather than fully functional.

7.6 Prediction and forgetting remain neutral

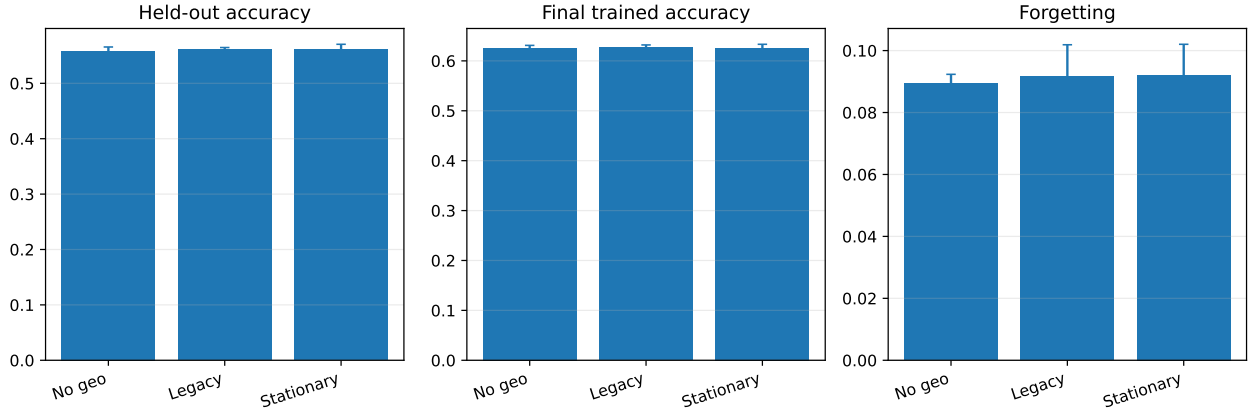


Figure 4: Operational metrics across five seeds. The small differences between methods are not statistically reliable.

Table 4: Aggregate operational results.

Method	Held-out accuracy	Final trained accuracy	Forgetting
No geometry	0.5583	0.6263	0.0894
Legacy full alignment	0.5612	0.6264	0.0916
Stationary full alignment	0.5607	0.6258	0.0922

Stationary full alignment changes held-out accuracy by only +0.23 percentage point, final trained accuracy by -0.05 point, and forgetting by +0.28 point. Every seed-level interval includes zero. C2 and C6 do not pass.

7.7 Causal specificity does not replicate

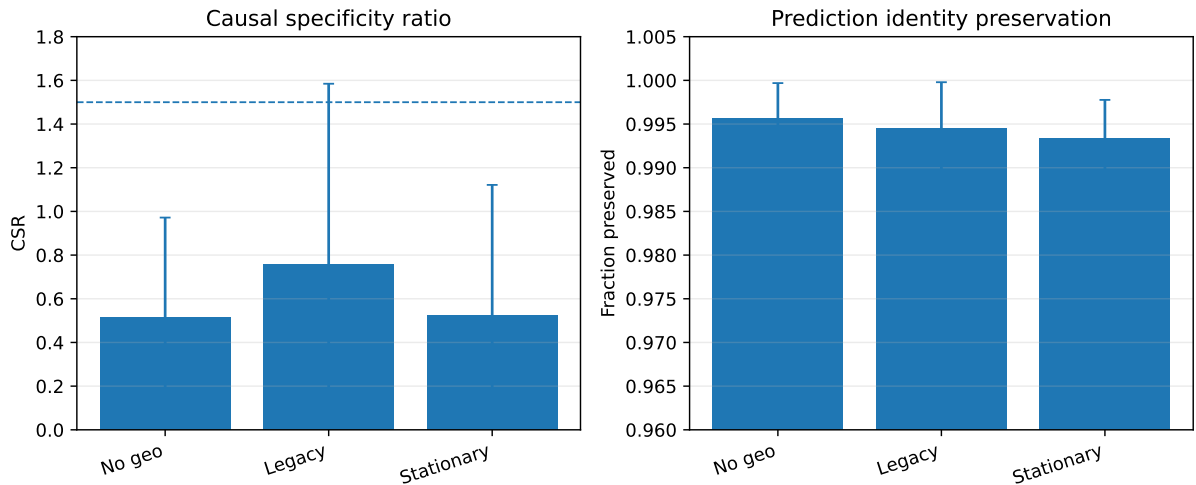


Figure 5: Causal specificity and prediction preservation. Prediction identity is preserved, but CSR remains below the candidate threshold and varies strongly across seeds.

The stationary model has mean CSR 0.523 with seed interval $[-0.075, 1.121]$, well below the 1.5 pilot threshold. Mean prediction identity preservation is high at 0.9934, and true-class log-probability

changes are small. Thus interventions are gentle but not factor-specific. The strong seed-0 v1.0.8 causal observation did not replicate.

7.8 Per-seed heterogeneity

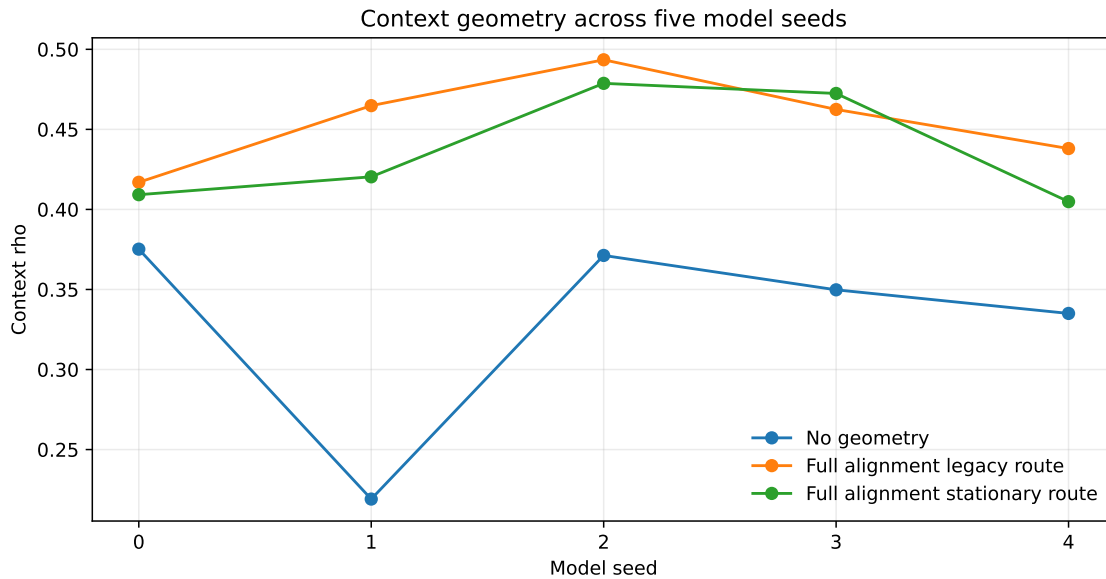


Figure 6: Context order by model seed. Full alignment improves context geometry in every seed, but effect magnitude varies substantially.

Five seeds are sufficient to identify a reproducible direction but not to estimate small effects precisely. Context-order improvement is robust in sign, while route, causal, and operational effects are dominated by between-seed variance.

8 Integrated interpretation

8.1 What v1.0.9 establishes

The pilot supports four narrow conclusions:

1. A context bottleneck can carry factor information when direct sensory shortcutting is prevented.
2. Four-dimensional local and cross-source alignment reproducibly improves context order and reduces context stress.
3. The learned context transfers factor information to unseen source identities.
4. The result is attributable to the complete context-alignment package rather than to the stationary route target alone.

8.2 What v1.0.9 falsifies or weakens

The pilot weakens several tempting interpretations:

- Better context geometry does not automatically produce better route geometry.
- Better context geometry does not automatically alter host synthesis.
- A theoretically cleaner route target need not produce a stronger empirical result.
- Single-seed causal specificity is not reliable evidence of C4.
- Geometric correlation alone is not an operational objective.

8.3 Representational geometry versus functional geometry

The evidence now requires a sharper vocabulary.

Representational geometry means that internal context distances encode a controlled factor, remain noncollapsed, and generalize across source identities.

Functional geometry would additionally require that this structure causally governs routing, host transformations, synthesis, memory transport, or operational performance.

v1.0.9 supports the former at candidate level and does not yet support the latter.

8.4 Why the result still matters

A replicated positive latent-geometry result is scientifically useful even without accuracy gain. It establishes that the architecture and loss family can create stable factor structure rather than a projection artifact. It also narrows the next problem: the bottleneck is no longer basic representability, but functional coupling.

The failed operational bridge is equally informative. It prevents overclaiming and suggests that future work should optimize mediation and transport, not merely increase ρ_{context} .

9 What is established, promising, failed, or untested

Claim	Status	Evidence and qualification
Executable matched-compute protocol	Established at development level	Five seeds completed; source and model seeds recorded; equal compute passed.
Context sufficiency under bottleneck	Strong development evidence	v1.0.7 interventions and competitive performance.
Replicated context geometry	Candidate pass	$\Delta\rho = 0.107$ with positive seed CI; stress decreases; all five seeds positive.
Held-out context factor decoding	Promising replicated evidence	$\Delta R^2 = 0.132$ and about 5° MAE reduction.
Stationary target superiority	Failed	Stationary-minus-legacy intervals include zero for primary geometry and operation metrics.
Replicated route geometry	Failed/inconclusive	Wide seed interval crossing zero.
Synthesis geometry	Failed/inconclusive	No improvement; stationary slightly worse than legacy.
Predictive utility	Failed/inconclusive	Accuracy intervals include zero.
Causal specificity	Failed	CSR below threshold and not replicated.
Host specialization	Untested in v1.0.9	Requires dedicated ablation and cross-seed role matching.
Dual-memory transport	Untested	Current anchor is controlled rehearsal, not MMALS dual memory.
Operational utility	Not qualified	No reliable benefit in accuracy, forgetting, cost, or stability.
Domain-general geometry	Untested	Only controlled RotatedMNIST has been studied.

10 Reviewer risk assessment

10.1 Risk 1: supervised shaping may be mistaken for discovery

The rotation angle is used in training losses. The paper must call the current protocol supervised geometric grounding. A later G1-B variant must exclude the factor from all losses before any claim of

unsupervised context discovery.

10.2 Risk 2: small number of model seeds

Five seeds are enough to reveal a replicated positive direction but not enough to stabilize small route or operational effects. Any submission should report all seed values, Student- t intervals, and positive-seed fractions rather than relying on a pooled sample-level p -value.

10.3 Risk 3: controlled synthetic transformation

Rotation provides unusually clean ground truth. The result may not transfer to overlapping classes, appearance drift, domain shift, or real temporal change. A second controlled factor and one non-synthetic domain are required before a general geometry claim.

10.4 Risk 4: the factor may be decodable but functionally ignored

The strongest current risk is that geometry is epiphenomenal. The classifier and host synthesis can solve the task without using the learned coordinate. Future protocols must explicitly test mediation and require function to depend on context.

10.5 Risk 5: legacy and stationary route targets are not cleanly separable in effect

Both full methods include the same context, fiber, and centroid terms. The target comparison is valid, but its effect is small relative to the broader alignment package. Claims must not attribute the replicated context result to the stationary target.

10.6 Risk 6: memory claims remain premature

The retention anchor uses old views already available in the paired batch. It is a controlled regularizer, not evidence for reconstructive or synthetic MMALS memory. C5 must remain unclaimed.

10.7 Risk 7: host ecology remains absent from the replicated pilot

A geometry paper about routes and hosts will face questions if host functions are not evaluated. Dedicated host contribution, replacement, locality, role matching, and seed stability remain necessary.

11 Publication positioning

11.1 What can be published now

Two publication forms are defensible.

Protocol and status paper. The full research history, evidence ladder, control logic, failure analysis, and v1.0.9 candidate result can support an arXiv technical report. Its central contribution would be methodological discipline for qualifying internal geometry in modular continual-learning systems.

Narrow empirical note. A shorter note could report the replicated context-order and held-out decoding result, explicitly framing it as supervised representational grounding on one controlled benchmark.

11.2 What is required for a stronger empirical G1 paper

A broader empirical paper should require:

1. a second factor or dataset;
2. an unsupervised or weakly supervised G1-B comparison;

3. explicit functional mediation from context to route and host behavior;
4. host specialization evidence across seeds;
5. continual transport and memory tests;
6. operational benefit or a clearly justified negative result;
7. frozen preregistered gates before qualification runs.

11.3 What should remain separate

Energy-guided routing belongs to G2 and phase-aware or density-state routing to G3. These extensions should not be merged into the current empirical claim. The order remains

$$\text{grounded geometry} \rightarrow \text{functional coupling} \rightarrow \text{energy-guided control} \rightarrow \text{phase-aware extensions.} \quad (18)$$

12 Recommended next experiments

12.1 Priority A: force functional use of context geometry

The next model should not merely make factor information available. It should require the route or host system to use it. Candidate mechanisms include:

- remove or severely regularize direct sensory-to-router access;
- predict route coordinates from context through a low-capacity map;
- add a mediation loss measuring route dependence on context;
- penalize successful prediction under context shuffling;
- require host-selection changes to follow the context tangent.

12.2 Priority B: couple geometry to host function

A route can be geometrically ordered while hosts compensate for it arbitrarily. Introduce transition consistency:

$$g_{r(u+\Delta u)}(z_0(u+\Delta u)) \approx \mathcal{T}_{\Delta u} g_{r(u)}(z_0(u)), \quad (19)$$

where $\mathcal{T}_{\Delta u}$ is a learned or known transport operator. This directly tests whether moving in context corresponds to a coherent functional transformation.

12.3 Priority C: intervention-trained geometry

Current causal probes are post hoc. A stronger design would train with local counterfactual consistency: perturb context along the learned factor tangent and require predictable route or synthesis movement while preserving class identity.

12.4 Priority D: memory transport

Replace direct old-view anchoring with bounded reconstructive and synthetic memory. Measure whether geometry, function, and predictions for old contexts can be reconstructed after later stages. This is the first legitimate C5 experiment.

12.5 Priority E: domain breadth

Replicate the same ladder on a second controlled factor such as scale, shear, illumination, or domain style, then on a real continual-learning dataset. The objective is not to maximize benchmarks immediately, but to test whether the geometry qualification method transfers.

12.6 Priority F: qualification design

Before the next multi-seed qualification run:

- freeze method variants and thresholds;
- separate pilot-tuned from qualification data;
- report all failed runs and numerical exceptions;
- retain exact source, model, and intervention seeds;
- map each manuscript claim to a CSV, figure, and gate.

13 Anticipated reviewer questions and direct answers

Did v1.0.9 prove that MMALS has a functional geometry?

No. It produced replicated candidate evidence for context representation geometry. Route, synthesis, causal specificity, and operational utility did not replicate.

Is the result just supervised metric learning?

The current G1-A protocol is supervised geometric grounding and should be described that way. The contribution is the controlled qualification of where the geometry appears and where it does not. Unsupervised discovery remains future work.

Why is the context result meaningful if accuracy is unchanged?

Because the result establishes a stable internal relation that generalizes across source identities. It narrows the engineering problem from representability to mediation. It is not yet an operational success.

Did the stationary route target work?

It worked numerically and removed curriculum-dependent target inconsistency. It did not outperform the legacy target empirically. The report treats this as a clean negative result.

Could the seed interval be optimistic with only five seeds?

Yes. The interval is a pilot estimate and should not be presented as final qualification. The consistency of sign across five seeds is more informative than a precise small-sample interval.

Why not move directly to energy or quantum-inspired routing?

Because the current geometry is not yet functionally coupled to routing and synthesis. Adding energy or phase structure now would multiply abstractions before the base causal chain is established.

What is the strongest reviewer-facing contribution now?

A falsifiable evidence ladder plus a replicated narrow result: globally aligned d4 training produces a more ordered and more decodable context space, while several stronger interpretations fail under matched controls.

14 Conclusion

Geometry-MMALS G1 has moved beyond conceptual metaphor and single-run visualization. The v1.0.9 pilot provides the first replicated candidate evidence that the system can learn a grounded latent context geometry on a controlled continual-learning benchmark. The effect survives model seeds and transfers to held-out source identities. Equally important, the same pilot shows that route geometry, synthesis geometry, prediction, forgetting, and causal specificity do not improve reliably, and that a theoretically cleaner stationary route target is not empirically superior.

The reviewer-facing conclusion should therefore be calibrated but substantive: the program has demonstrated representational context geometry, not yet functional geometry. The next scientific task is to make route and host function depend causally on that geometry, then test whether the resulting structure survives continual transport, domain variation, and operational constraints.

A Current gate table

Gate	Decision	Comment
C0 equal compute across five seeds	Pass	15,360 forwards and 80 optimizer steps per method and seed.
C1 stationary context ρ seed interval	Candidate pass	Mean effect +0.107, interval $[+0.029, +0.185]$.
C1 stationary context stress	Candidate pass	Mean effect -0.042 , interval $[-0.084, -0.001]$.
C1 route geometry	Fail/inconclusive	Interval crosses zero.
Stationary target beats legacy	Fail	Primary intervals cross zero.
Held-out factor decoding	Promising	R^2 and MAE improve across seeds.
C2 predictive superiority	Fail/inconclusive	Accuracy effects cross zero.
C3 host specialization	Not evaluated	Dedicated experiment required.
C4 source-bootstrap specificity	Fail	CSR below threshold.
C4 identity preservation	Candidate pass	Prediction preservation about 0.993.
C5 memory transport	Not implemented	Retention anchor is not dual memory.
C6 operational utility	Not qualified	No reliable accuracy, forgetting, or cost gain.

B Reproducibility snapshot

- Scientific version: Geometry-MMALS G1 v1.0.9.
- Build revision: `stationary-route-qualification-pilot-r3`.
- Model seeds: 0, 1, 2, 3, 4.
- Fixed source split and frozen sensory grove.
- Sensory acceptance accuracy: approximately 0.942.
- Equal compute: 15,360 image forwards and 80 optimizer steps per method per seed.
- Within-seed uncertainty: source-block bootstrap.
- Across-seed uncertainty: Student- t interval over model seeds.

- Dense held-out angle grid: every 15° from -75° to $+75^\circ$.
- Execution-stability tests: non-negative RNG seed mapping; collapsed and near-collapsed route backward tests.
- Archived evidence: aggregate tables, per-seed effects, causal summaries, run manifest, source hashes, and raw seed folders.

C Exact key evidence snapshot

Table 7: Stationary full alignment versus no geometry.

Evidence	Mean effect	95% lower	95% upper
Trained context ρ	+0.1071	+0.0289	+0.1852
Trained context stress	−0.0423	−0.0837	−0.0010
Trained route ρ	+0.0248	−0.0919	+0.1415
Trained synthesis ρ	−0.0064	−0.0246	+0.0119
Dense held-out context ρ	+0.0484	−0.0119	+0.1088
Held-out context R^2	+0.1323	+0.0930	+0.1716
Held-out angular MAE	−4.9889°	−6.3489°	−3.6289°
Held-out accuracy	+0.0023	interval crosses zero	−
Final trained accuracy	−0.0005	interval crosses zero	−
Forgetting	+0.0028	interval crosses zero	−

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